

RESEARCH ARTICLE

Fabrication and bending reliability evaluation of flexible electrowetting display device

Tinghong Yang, SID Student Member¹ | Miaoyang Wei¹ | Jianyang Guo² |
Caihong Sun¹ | Luyao Yan¹ | Xiaomeng Li¹ | Simin Ma¹ |
Dongxiang Luo³ | Rui Zhou¹ | Tang Biao¹ | Jiawei Lai¹  | Dong Yuan¹ 

¹Guangdong Provincial Key Laboratory of Optical Information Materials and Technology & Institute of Electronic Paper Displays, South China Academy of Advanced Optoelectronics, South China Normal University, Guangzhou, P. R. China

²Administrative and Research Office, South China Normal University, Guangzhou, China

³Huangpu Hydrogen Innovation Center/Guangzhou Key Laboratory for Clean Energy and Materials, School of Chemistry and Chemical Engineering, Guangzhou University, Guangzhou, China

Correspondence

Tang Biao, Lai Jiawei, and Dong Yuan
Guangdong Provincial Key Laboratory of
Optical Information Materials and
Technology & Institute of Electronic
Paper Displays, South China Academy of
Advanced Optoelectronics, South China
Normal University, Guangzhou 510006,
P. R. China.
Email: tangbiao@scnu.edu.cn; jiawei.lai@m.scnu.edu.cn; yuandong@scnu.edu.cn

Funding information

National Key R&D Program of China,
Grant/Award Number: 2023YFB3609400;
National Natural Science Foundation of
China, Grant/Award Numbers:
U23A20368, 12204181; Guangzhou Key
Laboratory of Electronic Paper Displays
Materials and Devices, Grant/Award
Number: 201705030007; Guangdong
Provincial Key Laboratory of Optical
Information Materials and Technology,
Grant/Award Number: 2023B1212060065;
Program for Guangdong Innovative and
Entrepreneurial Teams, Grant/Award
Number: 2019BT02C241; National Center
for International Research on Green
Optoelectronics, Grant/Award Number:
2016B01018; Natural Science Foundation
of Guangdong Province, Grant/Award
Numbers: 2024A1515011829,
2023A1515012962

Abstract

The Electrowetting Display (EWD) has numerous advantages making it a potential candidate for flexible displays. However, there has been limited research on the bending performance of Flexible Electrowetting Displays (FLEWD) and their feasibility in terms of flexibility has not been confirmed. To investigate the impact of bending on stability performance, FLEWD using flexible active-matrix substrate and PEN conductive substrates were fabricated. A systematic performance evaluation system for flexible devices is established. The results showed that FLEWD can display videos normally during bending. In the off state, the device can withstand a bending limit represented by a radius of 2 cm at a bending speed of 80 mm/s for 5,000 times. In the on state, the corresponding limits are 4 cm, 40 mm/s, and 100 times. Besides, the failure mode of FLEWD bending is clarified, which provides a guideline for the improvement of the flexural performance of flexible EWD in the future.

KEYWORDS

bending reliability, electronic paper display, Electrowetting display, flexibility threshold, flexible display

1 | INTRODUCTION

With the advancement of wearable and portable mobile devices, flexible displays have emerged as a promising research area due to their ultrathin, lightweight characteristics, exceptional mechanical flexibility, low power consumption, and extensive tunable emission capabilities.^{1,2} Various display technologies, such as Liquid Crystal Display (LCD),^{3,4} Organic Light Emitting Display (OLED),^{5,6} Electronic paper Display (E-paper),^{7,8} and Light-Emitting Diodes display (LED),^{9,10} have been extensively explored for their flexibility. Among these, electronic paper stands out as an excellent candidate for flexibility due to its low energy consumption and benefits for visual health. Flexible electronic paper demonstrations, including Electrophoretic Display (EPD),^{11–13} Electrochromic Display (ECD),^{14,15} and Electrowetting Display (EWD),^{16–18} have been developed, with EWD offering superior response time and flexibility due to its fluid-based working principle.

Given the advantages of electrowetting electronic paper, research on flexible electrowetting devices has increased. Early work demonstrated the feasibility of electrowetting on flexible substrates like paper, plastic, and metal.^{19,20} You et al.²¹ fabricated a Flexible Electrowetting Display (FLEWD) with a 45×21 pixel array, which exhibited normal operation even when bent. Hsieh et al.²² achieved a wide modulation range with lower saturation voltages on FLEWD using a flexible Polyethylene Glycol Terephthalate (PET) substrate. Li et al.^{23,24} enhanced bending performance and stability by using a Polyethylene Naphthalate (PEN) substrate and support columns. Qian et al.²⁵ addressed sealing failures and leakage issues with an optimization strategy for bending flexible devices. Recently, South China Normal University successfully illuminated a FLEWD prototype using a flexible ultrathin glass substrate, validating the feasibility of flexible EWD devices.

Although FLEWD devices have been successfully prepared and their structures optimized, their bending reliability has not been thoroughly investigated. Bending performance is crucial for the effective implementation of flexible electronic devices. Current criteria for bending performance evaluation are mainly suited for all-solid-state systems,^{26–31} which may not apply to EWD devices with flexible fluid materials. Therefore, a comprehensive study on the bending performance of liquid-based devices is needed.

This work establishes a bending reliability evaluation system and examines various bending parameters, such as bending radius, speed, and duration, for the FLEWD in both closed and open states. The study assesses the

electrical performance and oil movement within the pixel arrays after bending, validating the feasibility of FLEWD, providing insights into its flexural performance, and establishing a reliable evaluation method for bending reliability.

2 | EXPERIMENTAL SECTION

2.1 | FLEWD mechanism

The mechanism of FLEWD primarily relies on the principle of electrowetting, as shown in Figure 1A–C, governed by the Young-Lippmann equation (1).³²

$$\gamma_{LV} \cos \theta(V) = \gamma_{SV} - \gamma_{SL} + \frac{\epsilon_r \epsilon_0 V^2}{2d} \quad (1)$$

where γ_{SL} denotes the surface tension between the electrolyte and the conductor at zero electric field. γ_{SV} denotes the surface tension between the conductor and the external environment. γ_{LV} denotes the surface tension between the electrolyte and the external environment. θ denotes the contact angle between the electrolyte and the dielectric. $\epsilon_r \epsilon_0 / d$ denotes the capacitance of the interface for a uniform dielectric of thickness d and dielectric constant ϵ_r . V denotes the effective applied voltage.

Recent studies have shown the effect of surface curvature on electrowetting can be described by the following equations³³:

$$\begin{aligned} \text{Convex cylindrical surface : } \cos \theta(V) \\ = \cos \theta_0 + \frac{\xi V^2}{2\gamma_{LV}} \frac{1}{(R+d) \ln \left(\frac{R+d}{R} \right)} \end{aligned} \quad (2)$$

$$\begin{aligned} \text{Concave cylindrical surface : } \cos \theta(V) \\ = \cos \theta_0 + \frac{\xi V^2}{2\gamma_{LV}} \frac{1}{(R-d) \ln \left(\frac{R}{R-d} \right)} \end{aligned} \quad (3)$$

where R is the radius of the surface and d is the thickness of the dielectric layer. The influence of the curved surface electrowetting effect can be determined by the ratio of the thickness of the dielectric layer to the radius of curvature. So when the radius of curvature is greater than 1 mm, the effect of curvature on the contact angle can be neglected, with an angle deviation of less than 0.05° .

The EWD device comprises upper and lower substrates, electrodes, polar liquid (generally deionized

water), hydrophobic insulating layer, pixel wall array, oil, and encapsulating adhesive. When no voltage is applied, the device remains closed. Upon voltage application, the contact angle between oil and hydrophobic insulating layer changes, causing the device to open, as shown in Figure 1D.

2.2 | Materials and fabrication

In an EWD device, the polar liquid, hydrophobic insulating layer, pixel wall array, oil, and encapsulation adhesive are all flexible materials. Consequently, the flexibility of the EWD device largely depends on the top and bottom substrates. Common flexible substrates include paper, ultra-thin glass, polymer, and metal. The following presents examples of the fabrication process of FLEWD devices utilizing PEN substrates and ultra-thin glass substrates.

The fabrication of FLEWD with PEN as a substrate involves five main steps: bottom substrate preparation, top substrate preparation, filling, encapsulation, and stripping of the flexible device. This process is detailed and illustrated in Figure 1E. Initially, the process begins with preparing a bottom substrate consisting of glass and

PEN with an indium tin oxide (ITO) electrode. A hydrophobic insulating material, Hyflon AD 40H (Solvay), is then coated onto the ITO substrate, with a thickness of about 800 nm. Subsequently, a photoresist material is applied, and photolithography is used to fabricate a continuous pixel wall structure.³⁴ For the top substrate, the support pillar structure is fabricated on the PEN with ITO. This structure prevents direct contact between the top and bottom substrates during bending and minimizes the risk of oil spillage from the pixel walls. Next, oil filling is performed using self-assembly.³⁵ Following filling, the bottom and top substrates are laminated by encapsulating the device in polar liquid. Finally, After the device is press-fitted and dried for 48 hours, the glass substrate is removed, completing the FLEWD fabrication.

For FLEWDs based on ultra-thin glass TFT (UTG-TFT) substrates, the process omits the steps of adhesion and peeling from the glass. Other steps are similar to those used for PEN-based FLEWDs.

2.3 | Bending test methods

The bending test platform used in this work is shown in Figure 2A. The FLEWD devices with a PEN substrate

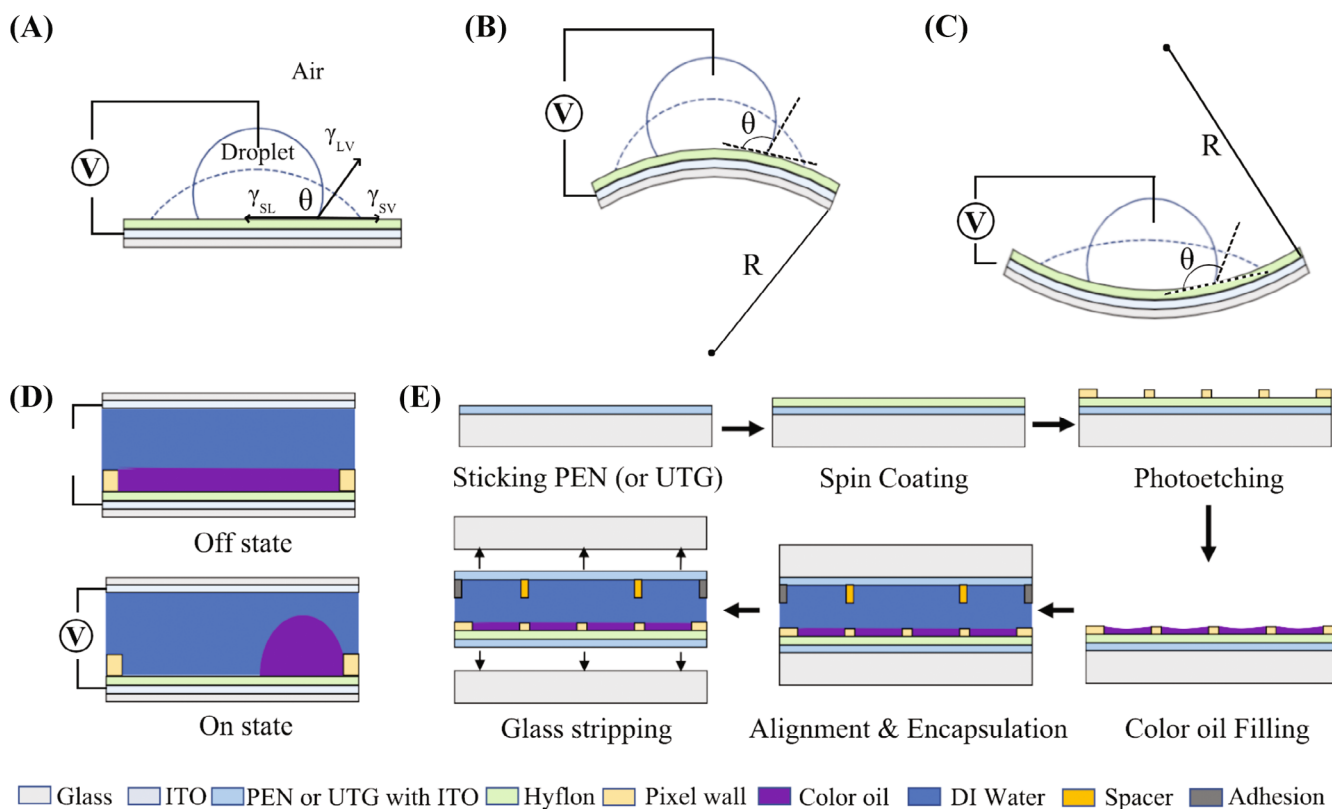


FIGURE 1 Schematic diagram of the FLEWD principle and preparation process, (A) electrowetting on flat surface; (B) electrowetting on a convex surface; (C) electrowetting on a concave surface; (D) electrowetting display at off state and on state; (E) preparation process of FLEWD with PEN (UTG) as a substrate.

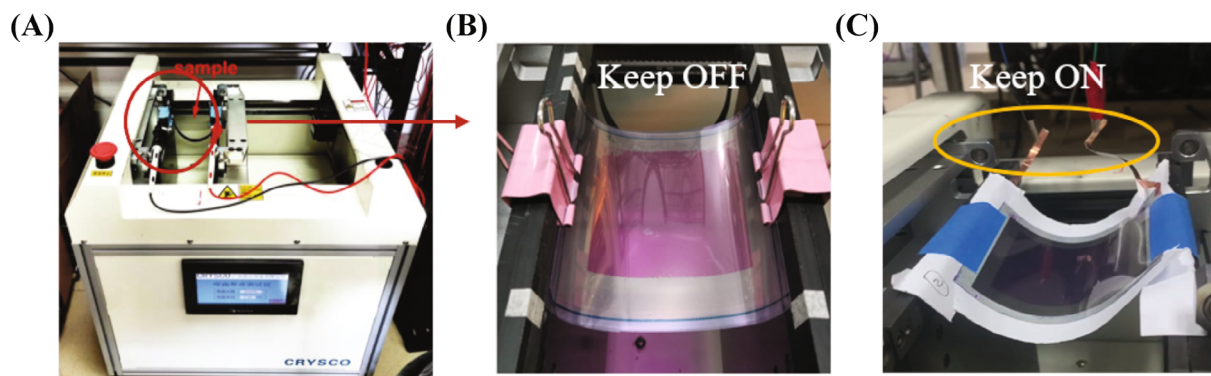


FIGURE 2 Bending test for FLEWD (A) Equipment; (B) off-state device; (C) on-state device.

(size: 10 cm × 10 cm) were secured on the platform, and bending experiments were performed by adjusting various parameters. For the bending radius tests, radii ranging from 2 cm to 5 cm were used at a constant speed of 10 mm/s to minimize speed-related impacts. In bending speed tests, velocities from 10 mm/s to 80 mm/s were applied with a fixed radius of 5 cm to control for radius effects. Bending count tests involved counts from 1 to 10,000, maintaining a velocity of 10 mm/s and a radius of 5 cm.

Bending tests were conducted in two states: off state with no driving voltage, as shown in Figure 2B, and open state with a 25 V applied voltage, as shown in Figure 2C. During and after bending, electrical performance was assessed using an impedance analyzer (Wayneker, R6500B), measuring capacitance-voltage (C-V) and leakage current-voltage (I-V) curves with an applied voltage ranging from 0 V to 30 V, 2 V per step. The response time was measured by monitoring luminance changes using a colorimeter (Admesy, ARGES45). Additionally, oil distribution within the pixel array was inspected under a microscope (Hours, CSUC-200C).

3 | RESULT AND DISCUSSION

Fabrication of ELEWD devices, capable of image and video display, was successfully accomplished. These devices continued to operate effectively even with a bending radius of 5 cm, as demonstrated in Figure 3A. A comprehensive parameter analysis was then performed to evaluate the device's reliability. Since electrowetting display devices can be regarded as capacitors, they have a correlation with voltage. In other words, the capacitance versus voltage (C-V) profile reflects the device's switching performance.³⁶ So, when the device fails, noticeable deviations appear in the C-V curve. Three rigorous tests at 5 cm bending radius revealed very small differences in the C-V

curve, demonstrating the device's stability at this bending state, as shown in Figure 3B. Additionally, the device exhibited a response time of 30 ms where the response time Δt is defined as $\Delta t = t_{on} - t_{off}$ at the same bending radius, as shown in Figure 3C, consistent with previously reported EWD response times.³⁷ The device also achieved four levels of grayscale, as shown in Figure 3D, highlighting its substantial practical and application potential. Given the inherent low reliability of flexible UTG TFTs, and to isolate the impact of the substrate on the bending reliability of FLEWD, the upcoming bending tests will employ FLEWD on a PEN substrate.

A failure in the dielectric layer can lead to a leakage current increase within the device.³⁸ To evaluate this, leakage current-voltage (I-V) curves were measured. Even under conditions involving a bending radius of 5 cm, a bending speed of 80 mm/s, and 10,000 bending cycles, the variation in the device's leakage current remains very small, as exemplified in Figure 4A-C. These results indicated the leakage current varies in the hundreds of nA range, thus indirectly affirming the relative stability of the dielectric layer properties after bending. This implies that the bending assessments conducted within this study did not compromise the dielectric layer properties of the device.

The C-V and response time curves for various off-state bending tests are shown in Figure 5, with microscopic images of the pixel areas after bending in the inset. The analysis indicates that the C-V curves remain almost unchanged from the initial unbent state, even with a bending radius of 2 cm, as depicted in Figure 5A. This stability is reflected in the consistent turn-on voltage and hysteresis effects. Additionally, oil distribution in the pixels before and after bending showed no defects like overflow or support pillar collapse, affirming the device's structural integrity. The response time curves also remained stable, further supporting the device's reliability as depicted in Figure 5B.

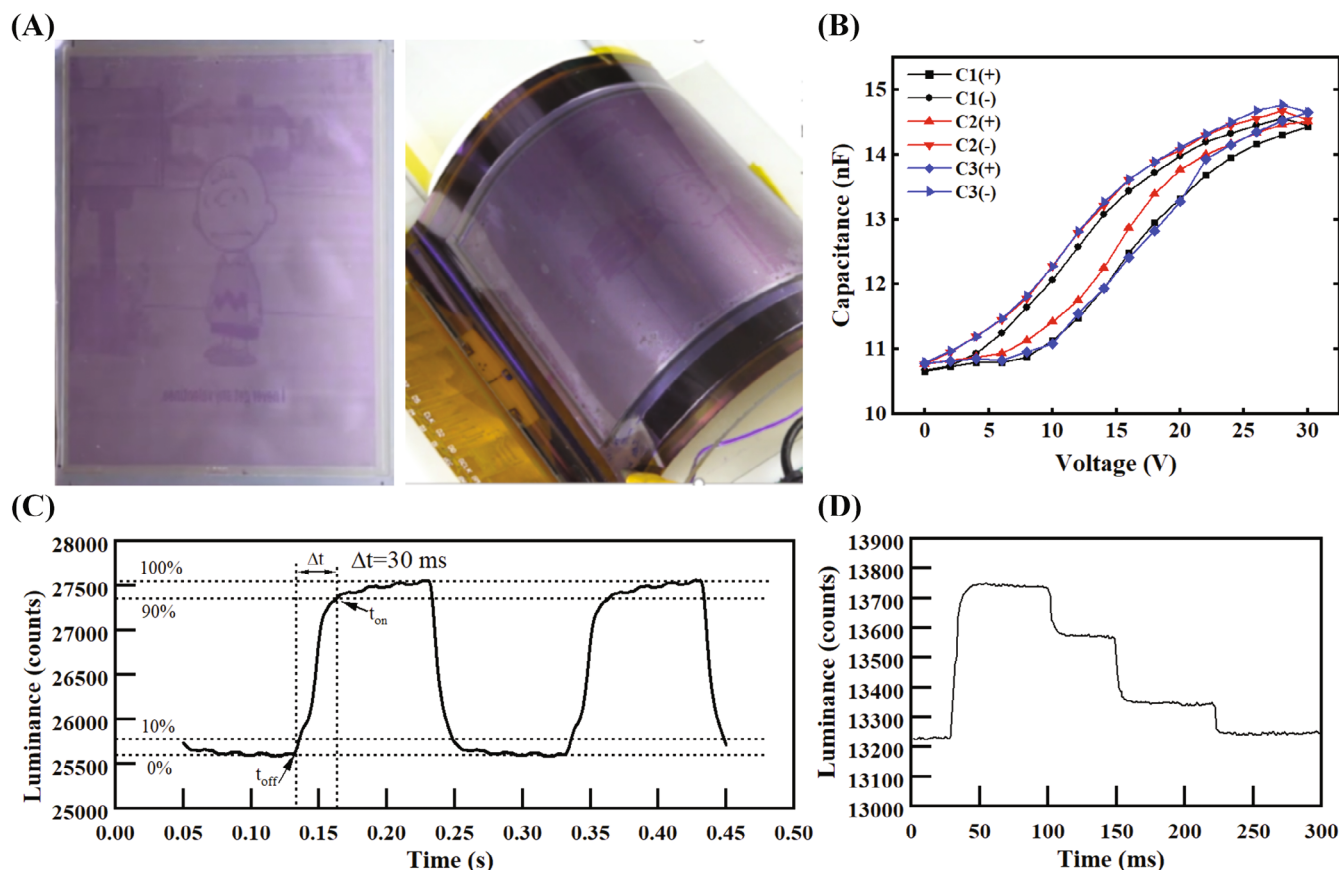


FIGURE 3 Basics performance of FLEWD (A) prototype demonstration; (B) Capacitance-Voltage curves; (C) Response time curves; (d) Gray level.

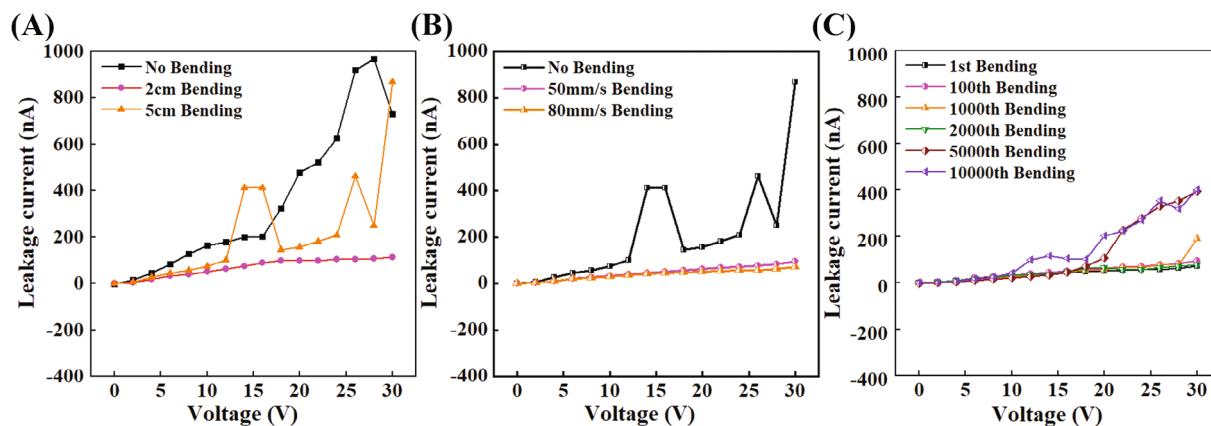


FIGURE 4 The Leakage current-Voltage curves after bending test under different bending conditions, (A) various bending radii; (B) various bending velocities; (C) various bending times.

Tests at varying bending speeds, presented in Figure 5C–D, demonstrate that even at 80 mm/s, the C-V curves and response time curves remain stable, indicating that bending speed does not affect oil stability in the off-state. Iterative bending tests, depicted in Figure 5E, show stability up to 5,000 bends. However, after 10,000 bends, there is a noticeable shift in the C-V curve and an

increase in initial capacitance, attributed to changes in oil distribution and overflow in some pixels as shown in the insets of Figure 5E. This shift affects the response time, which increases after 10,000 bends are illustrated in Figure 5F. Despite no dielectric failure observed in leakage current tests, the overflow likely results from “fatigue” which may lead to changes in the

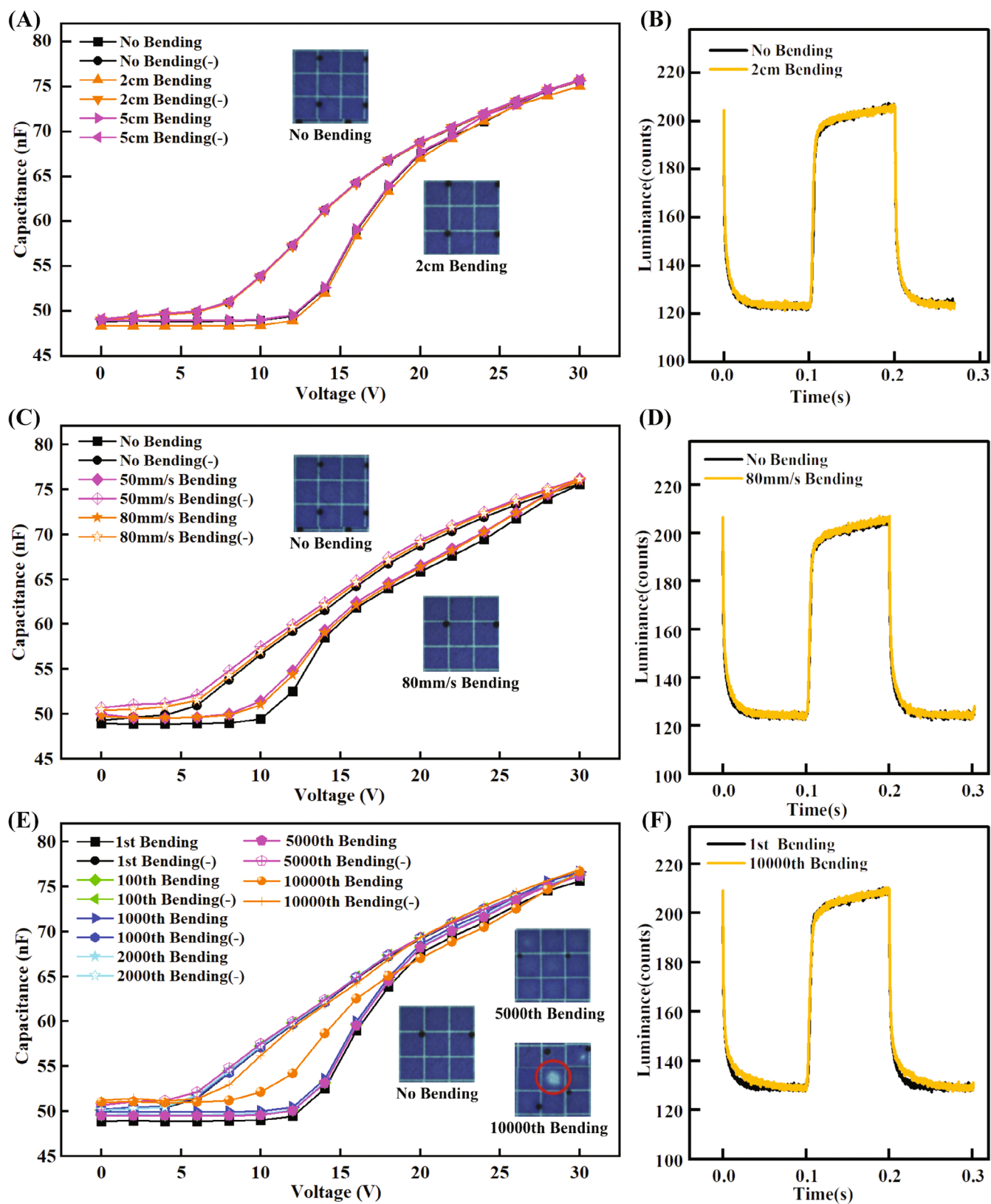


FIGURE 5 Capacitance-Voltage curves and Response time curves of devices after bending at off-state, (A) and (B) various bending radii; (C) and (D) various bending velocities; (E) and (F) various bending times. The Insert is the photograph of pixels observed under the microscope and the black dots we can see is the support pillars.

microstructure within the material by repeated bending or deformation in the pixel wall affecting oil wettability. To address this issue, a thorough evaluation of pixel walls and material optimization is recommended.

The photoelectric response properties of devices bent in the on state were also studied in detail. Firstly, the capacitance under different bending conditions was measured, providing indirect insight into the contraction status of the oil in pixels while the device is on, as shown in Figure 6. As the bending radius decreases, the capacitance remains around 69 nF, indicating no significant change in oil contraction with varying bending radii, as depicted in Figure 6A. Additionally, bending speed does not impact the oil's contracted state in the on state, as illustrated in Figure 6B. However, capacitance gradually decreases with increasing bending cycles, stabilizing after approximately 1,000 bends, as exhibited in Figure 6C. This decrease suggests that, while the device is on, the oil in the pixel gradually spreads back during repeated bending, eventually reaching a steady state. This phenomenon that the oil may not maintain its desired configuration within the pixel resulting in backflow, known as "oil backflow",³⁹ is characterized by reduced capacitance due to the oil's re-spreading. Furthermore, charge trapping in EWD devices will reduce charge accumulation at the interface between the liquid and dielectric layers. This further reduces the electrowetting force at the three-phase contact line (oil/water/dielectric layer), leading to a decreased contact angle.³⁶ Thus, the increased number of bends affects the oil contact angle within the pixel, which in turn impacts the device's display stability.

The C-V and response time curves for different on-state bending tests are shown in Figure 7, with an inset featuring microscopic images of device areas after bending. Analysis of the C-V curves reveals that the capacitance does not revert to its initial value at a bending

radius of 3 cm. Pixel morphology at this radius shows the oil flipping over the pixel wall, as shown in Figure 7A. Concurrently, the response time curve starts to deviate from its conventional value at the same radius, as depicted in Figure 7B. This indicates that smaller bending radii hinder reversibility under a charged condition. This issue arises because, in the open state, the initially flat oil contracts toward the pixel wall corner, increasing vertical elevation. When subjected to a bending radius, the oil's tendency to flip over the wall is heightened, leading to failure. Optimizing the pixel wall structure with a higher design could address this issue, allowing for a larger bending radius.

The reversibility deteriorates after a bending velocity of 50 mm/s, as shown in Figure 7C. Pixel morphology reveals that at 40 mm/s, the oil spreads evenly within the pixel grid without overflow. However, at 50 mm/s and higher, oil overflow becomes apparent. Similarly, as shown in Figure 7D, response time curves exhibit significant deviations at bending speeds of 50 mm/s and above. Rapid bending induces shear flow within the device's fluid, and the force from this shear flow increases with speed, potentially leading to oil overflow. For instance, at 80 mm/s, approximately 70% of the pixel grid area experiences oil overflow.

In the bending cycle test, significant shifts in the C-V curve occur after 1,000 bends, disrupting the typical capacitance-voltage relationship, as shown in Figure 7E. Initial bending (up to 100 bends) shows no change in oil distribution condition, but as the number of bends increases, oil overflow becomes more pronounced, particularly after 1,000 bends. This results in substantial oil agglomeration and eventual shedding of the support pillar. Correspondingly, the response time curve shows noticeable deviations after 1,000 bends, as depicted in Figure 7F. This could be due to pixel wall "fatigue" from

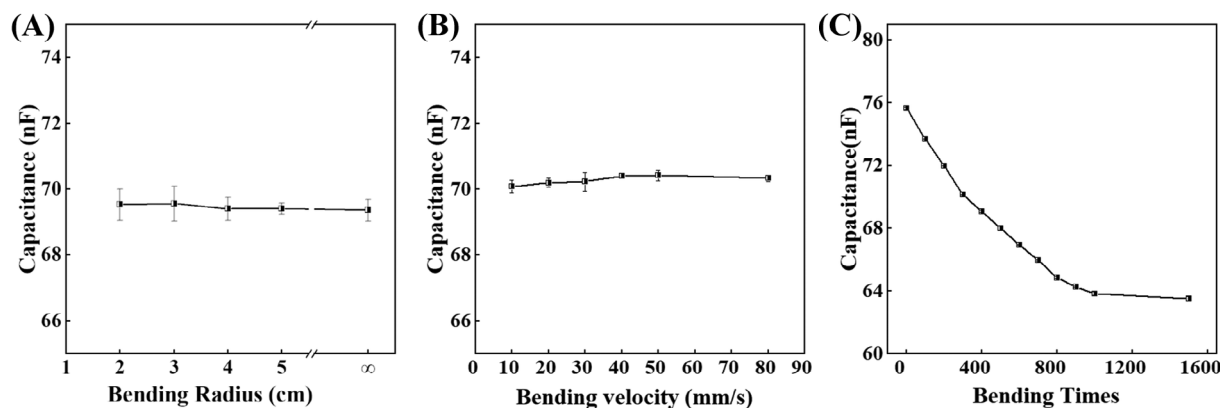


FIGURE 6 Capacitance measurement under different bending conditions of on-state devices, (A) various bending radii; (B) various bending velocities; (C) various bending times.

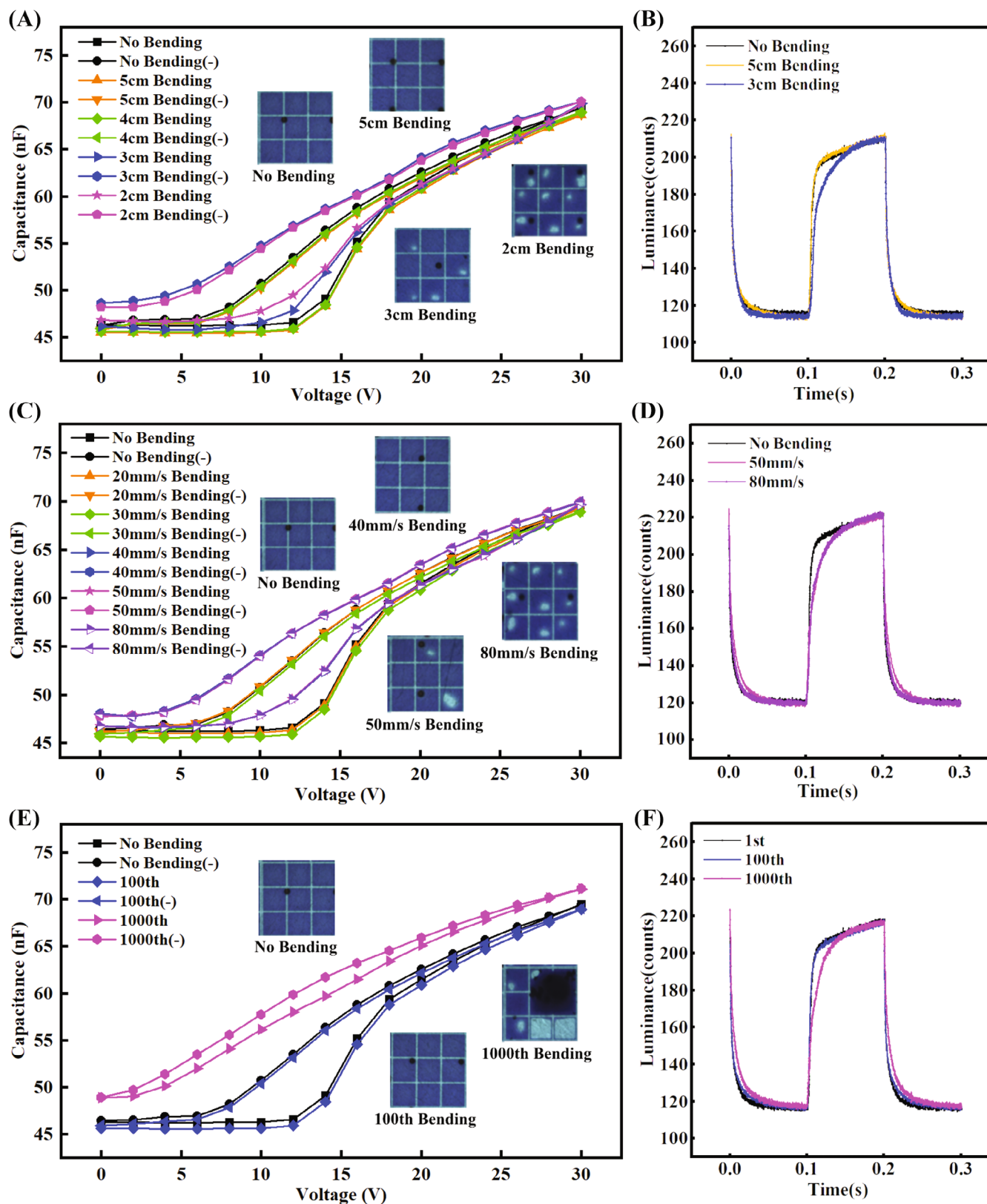


FIGURE 7 Capacitance-Voltage curves and Response time curves after bending test under various conditions of on-state devices, including (A-B) various bending radii; (C-D) various bending velocities; (E-F) various bending times. The Insert is the photograph observed under the microscope and the black dots we can see is the support pillars.

repeated bending which enhanced wettability with oil, and the increase of oil's vertical height in the open state, causing device failure after fewer bends.

In the open state, devices exhibit failure under various bending conditions, particularly due to oil flipping over walls. While the structural integrity of other functional layers remains intact, enhancing pixel wall design could improve reliability under different bending conditions. Implementing elevated pixel wall configurations may help bolster device reliability across various bending scenarios.

Generally, the leakage current values of the FLEWD device remain remarkably consistent across various bending conditions, both in on and off states. This consistency highlights the robust stability of the device's dielectric layer throughout testing. In the off state, factors such as the bending radius (≤ 2 cm), bending speed (≤ 80 mm/s), and bend count ($\leq 5,000$ times) have minimal impact on the device's electrical properties and oil condition. Similarly, in the on state, parameters including bending radius (≥ 4 cm), bending speed (≤ 40 mm/s), and bend count (≤ 100) show negligible effects on electrical characteristics and oil state. These findings establish a benchmark for assessing the flexibility and performance of FLEWD devices, forming a comprehensive evaluation framework.

4 | CONCLUSION

In summary, FLEWD was prepared using an efficient fabrication process. The performance of FLEWDs in both the off-state and on-state was tested under various bending conditions. By analyzing electrical properties and oil distribution in the pixel grid after bending, we determined the safe bending limits for FLEWDs. This assessment validates the bending capabilities of FLEWD devices and provides crucial benchmarks for their evaluation.

The primary failure modes identified were oil overflow and inability to return to the original state. Future research should focus on optimizing materials, refining pixel structure design, and stabilizing the oil state to enhance bending stability. Additionally, further theoretical studies on bending failure mechanisms are recommended.

Overall, this research is significant for guiding the optimization of FLEWD device structures and manufacturing processes.

ACKNOWLEDGMENTS

This work was supported by the National Key R&D Program of China (No. 2023YFB3609400), the National Natural Science Foundation of China (No. U23A20368,

No. 12204181), the Natural Science Foundation of Guangdong Province (No. 2024A1515011829, No. 2023A1515012962), the Program for Guangdong Innovative and Entrepreneurial Teams (No. 2019BT02C241), Guangdong Provincial Key Laboratory of Optical Information Materials and Technology (No. 2023B1212060065), Guangzhou Key Laboratory of Electronic Paper Displays Materials and Devices (No. 201705030007), National Center for International Research on Green Optoelectronics (No. 2016B01018), MOE International Laboratory for Optical Information Technologies and the 111 Project.

ORCID

Jiawei Lai  <https://orcid.org/0000-0002-1351-4268>

Dong Yuan  <https://orcid.org/0000-0002-5446-5570>

REFERENCES

1. Zhang DD, Huang TY, Duan L. Emerging self-emissive Technologies for Flexible Displays. *Adv Mater*. 2020;32(15):1902391. <https://doi.org/10.1002/adma.201902391>
2. Raupp GB. Flexbbible Display Center at Arizona State University: A Unique Industry-Government-Academic Partnership for Creating Revolutionary Information Display Technology. *IEEE* 2008; 31–3.
3. Vogels J, Klink SI, Penterman R, et al. Robust flexible LCDs with paintable technology. *J Soc Inf Disp*. 2004;12(4):411–6. <https://doi.org/10.1889/1.1847740>
4. Xie M, Qin W, Kong X, Liu J, Qin F. Flexible reflective liquid crystal display technology. *SID symposium digest of technical papers. SID Symp Digest Technical Papers*. 2021;52(1):1048–51. <https://doi.org/10.1002/sdtp.14871>
5. Tao J, Wang R, Yu H, Chen L, Fang D, Tian Y, et al. Highly transparent, highly thermally stable Nanocellulose/polymer hybrid substrates for flexible OLED devices. *ACS Appl Mater Interfaces*. 2020;12(8):9701–9. <https://doi.org/10.1021/acsami.0c01048>
6. Liu YF, Feng J, Bi YG, Yin D, Sun HB. Recent developments in flexible organic light-emitting devices. *Adv Mater Technol*. 2019;4(1):1800371. <https://doi.org/10.1002/admt.201800371>
7. Heikenfeld J, Drzaic P, Yeo J-S, Koch T. Review paper: a critical review of the present and future prospects for electronic paper. *J Soc Inf Disp*. 2021;19(2):129–56. <https://doi.org/10.1889/JSID19.2.129>
8. Chen Y, Au J, Kazlas P, Ritenour A, Gates H, McCreary M. Flexible active-matrix electronic ink display. *Nature*. 2003; 423(6936):136. <https://doi.org/10.1038/423136a>
9. Zhao J, Lo LW, Wan H, Mao P, Yu Z, Wang C. High-speed fabrication of all-inkjet-printed organometallic halide perovskite light-emitting diodes on elastic substrates. *Adv Mater*. 2021; 33(48):2102095. <https://doi.org/10.1002/adma.202102095>
10. Chuang T-K, Troccoli M, Kuo P-C, et al. Top-emitting 230 dots/in. Active-matrix polymer light-emitting diode displays on flexible metal foil substrates. *Appl Phys Lett*. 2007;90(15):151114.
11. Venugopal SM, Allee DR. Integrated a-Si:H source drivers for 4" QVGA electrophoretic display on flexible stainless steel substrate. *IEEE J Display Technol*. 2007;3(1):57–63. <https://doi.org/10.1109/JDT.2006.890712>

12. Sun JF, Sapkota A, Park H, Wesley P, Jung Y, Maskey BB, et al. Fully R2R-printed carbon-nanotube-based limitless length of flexible active-matrix for electrophoretic display application. *Adv Electron Mater*. 2020;6(4):1901431. <https://doi.org/10.1002/aelm.201901431>
13. Park CB, Kim KM, Lee JE, Na HI, Yoo SS, Yang MS. Flexible electrophoretic display driven by solution-processed organic TFT with highly stable bending feature. *Org Electron*. 2014;15(12):3538–45. <https://doi.org/10.1016/j.orgel.2014.09.039>
14. Chul MH, Chang-Hyun K, Lodge TP, et al. Multicolored, low-power, flexible electrochromic devices based on ion gels. *ACS Appl Mater Interfaces*. 2016;32(2):6252–60.
15. Dubey A, Tao X, Cochrane C, Koncar V. Textile based three-layer robust flexible and stable electrochromic display. *IEEE Access*. 2020;8:182918–29. <https://doi.org/10.1109/ACCESS.2020.3028655>
16. Hayes RA, Feenstra BJ. Video-speed electronic paper based on electrowetting. *Nature*. 2003;425(6956):383–5. <https://doi.org/10.1038/nature01988>
17. Lao Y, Sun B, Zhou K, et al. Ultra-high transmission electrowetting displays enabled by integrated reflectors. *IEEE J Display Technol*. 2008;4(2):120–2. <https://doi.org/10.1109/JDT.2008.920172>
18. Heikenfeld J, Steckl AJ. High-transmission electrowetting light valves. *Appl Phys Lett*. 2005;86(15):34–40. <https://doi.org/10.1063/1.1901816>
19. Kim DY, Steckl AJ. Electrowetting on paper for electronic paper display. *ACS Appl Mater Interfaces*. 2010;2(11):3318–23. <https://doi.org/10.1021/am100757g>
20. You H, Steckl AJ. Electrowetting on flexible substrates. *J Adhes Sci Technol*. 2012;26(12):1931–9. <https://doi.org/10.1163/156856111X600244>
21. You H, Steckl AJ. Versatile electrowetting arrays for smart window applications-from small to large pixels on fixed and flexible substrates. *Sol Energy Mater sol Cells*. 2013;117(6):544–8. <https://doi.org/10.1016/j.solmat.2013.07.027>
22. Hsieh IC, Chen LY, Chen HY, et al. Low driving voltage of electrowetting display on flexible substrate. In: *Flexible electronics and displays conference and exhibition*; 2008. p. 1–5.
23. Li H, Zhao W, Jiang H, Yue Q, Tang B, Dou Y, et al. The fabrication of flexible Electrofluidic display arrays based on PEN substrate. *SID Symp Digest Technical Papers*. 2018;49(5):115–7. <https://doi.org/10.1002/sdtp.12656>
24. Li H, Li RH, Jiang HW, et al. Fabrication and evaluation of flexible electrowetting display with support pillars. *Nanoscale Adv*. 2020;2(9):4077–84. <https://doi.org/10.1039/D0NA00355G>
25. Qian Y, Wang T, Wei M, Tang B, Zhou G. Reliability studies on multilayer thin-film materials and seal architecture for flexible Electrofluidic displays under bending stress[C]. *SID Symp Digest Technical Papers*. 2020;51(6):104–8. <https://doi.org/10.1002/sdtp.13764>
26. Grego S, Lewis J, Vick E, Temple D. Development and evaluation of bend-testing techniques for flexible-display applications. *J Soc Inf Disp*. 2012;13(7):575–81. <https://doi.org/10.1889/1.2001215>
27. Tabares G, Mhedhbi S, Lesecq M, Damilano B, Brault J, Chenot S, et al. Impact of the bending on the electroluminescence of flexible InGa_N/Ga_N light-emitting diodes. *IEEE Photonics Technol Lett*. 2016;28(15):1661–4. <https://doi.org/10.1109/LPT.2016.2563258>
28. Chiang CJ, et al. Mechanical modeling of flexible OLED devices. *Org Electron*. 2009;10(7):1268–74.
29. Chiang CJ, Winscom C, Monkman A. Electroluminescence characterization of FOLED devices under two type of external stresses caused by bending. *Org Electron*. 2010;11(11):1870–5. <https://doi.org/10.1016/j.orgel.2010.08.021>
30. Zhou HY, Park J-W. Elucidating the influences of mechanical bending on charge transport at the interfaces of organic light-emitting diodes. *Thin Solid Films*. 2016;619:281–7. <https://doi.org/10.1016/j.tsf.2016.10.050>
31. Niu L, Ding J, Liu W. Analysis on the mechanical behavior of flexible screens. *Materials*. 2022;15(8):2829. <https://doi.org/10.3390/ma15082829>
32. Shamaï R, Andelman D, Berge B, Hayes R. Water, electricity, and between ... on electrowetting and its applications. *Soft Matter*. 2008;4(1):38–45. <https://doi.org/10.1039/B714994H>
33. Wang Y, Zhao Y-P. Electrowetting on curved surfaces. *Soft Matter*. 2012;8(9):2599–606. <https://doi.org/10.1039/c2sm06878h>
34. Tang B, Wei MY, Shao W, Barman J, Sun H, Lu L, et al. Facile fabrication of binary wettability patterned microstructure for microfluidics. *J Micromech Microeng*. 2021;31(4):045007. <https://doi.org/10.1088/1361-6439/abe20a>
35. Liquavista B V, Method of manufacturing an optical display: US 2012/0127555 A1 [P]. 2012-05-20.
36. Kim DY, Steckl AJ. Complementary electrowetting devices on plasma-treated fluoropolymer surfaces. *Langmuir*. 2010;26(12):9474–83.
37. Guo Y, Tang B, Yuan D, Bai P, Li H, Yi Z, et al. Invited paper: Electrowetting display: towards full-color video reflective display. *SID Symp Digest Technical Papers*. 2021;52(2):59–63. <https://doi.org/10.1002/sdtp.15020>
38. Zhou R, Ye QL, Li H, Jiang H, Tang B, Zhou G. Experimental study on the reliability of water/fluoropolymer/ITO contact in electrowetting displays. *Results Phys*. 2019;12(2):1991–8. <https://doi.org/10.1016/j.rinp.2019.02.037>
39. Jiang C, Tang B, Xu B, Groenewold J, Zhou G. Oil conductivity, electric-field-induced interfacial charge effects, and their influence on the electro-optical response of electrowetting display devices. *Micromachines*. 2020;11(7):702–11. <https://doi.org/10.3390/mi11070702>

AUTHOR BIOGRAPHIES



Tinghong Yang is a doctoral student of South China Academy of Advanced Optoelectronics, South China Normal University. His research interest includes electrowetting display.



Miaoyang Wei is a graduate student of South China Normal University. Now, she is working in Hikvision Technology Co. Ltd. on smart sensor algorithms.



Jianyang Guo graduated from South China Normal University (SCNU) with a master's degree in 2013, majoring in optical engineering, and has been working at SCNU since 2018, specializing in laser-related research.



Caihong Sun graduated from South China Academy of Advanced Optoelectronics, South China Normal University with a master's degree. Her research interest includes electrowetting display.



Luyao Yan studied in South China Academy of Advanced Optoelectronics, South China Normal University. Her research interest includes electrowetting display.



Xiaomeng Li studied in South China Academy of Advanced Optoelectronics, South China Normal University. Her research interest includes electrowetting display.



Simin Ma studied in South China Academy of Advanced Optoelectronics at South China Normal University. Her research interest includes electrowetting display.



Dongxiang Luo male, Ph.D., Associate Professor, obtained his doctoral degree from South China University of Technology in 2014. His primary research focus is on semiconductor optoelectronic materials and devices.



Dr. Rui Zhou received his BS degree from the School of Mechanical and Automotive Engineering, South China University of Technology (SCUT), China, from 2005 to 2009; From 2009 to 2014, he received his PhD degree from the School of Mechanical and Automotive Engineering, South China University of Technology, and from September 2012 to September 2013, he received joint training at the S.M. Wu Manufacturing Research Center of the University of Michigan, USA. In 2015, he joined the Institute of Color Dynamic E-paper Display Technology, South China Institute of Advanced Photonics, South China Normal University. Currently, he is mainly engaged in the development of new electrowetting display technology and flexible sensor technology based on printing technology, especially the inkjet printing technology of new functional materials and the fabrication of surface functional structures.



Dr. Tang Biao graduated from Anhui University in 2005, received his PhD degree from South China University of Technology in 2013, and joined South China Academy of Advanced Optoelectronics, South China Normal University in the same year, focusing on the innovative research and development of color and dynamic e-paper based on the principle of electrowetting display for a long time.



Dr. Jiawei Lai received PhD degree from the Magnetic Materials and Functional Thin Films Team of the South China University of Technology in 2018. He worked as a joint PhD from 2015 to 2018 at the Delft University of Technology, the Netherlands, and worked as a postdoctoral researcher from 2019 to 2022 at the National Institute for Materials Research, Japan. In 2022, joined the South China Academy of Advanced Optoelectronics, South China Normal University, and is engaged in the direction of color dynamic electronic paper display materials and devices.



Dr. Dong Yuan is a professor at the South China Academy of Advanced Optoelectronics, South China Normal University. She received her PhD in mechanical manufacturing and automation from South China University of Technology in 2013.

Since 2014, she has been conducting research on optoelectronic materials and devices at the South China Academy of Advanced Optoelectronics. Currently, she serves as the principal investigator of the Young Scientists Program under the National Key Research and

Development Program of China, focusing on color and flexible electronic paper display materials and devices.

How to cite this article: Yang T, Wei M, Guo J, Sun C, Yan L, Li X, et al. Fabrication and bending reliability evaluation of flexible electrowetting display device. *J Soc Inf Display*. 2025;33(7): 861–72. <https://doi.org/10.1002/jsid.2085>